

Application of Sequences: Interest, Appreciation, Depreciation, etc.

Introduction: When Money Follows a Pattern

Without realizing it, people make a lot of financial decisions that rely on mathematical patterns. When a family saves for a child's college education, a business owner calculates the future value of an asset, or a borrower takes out a loan to buy a house, they are all interacting with sequences and series. These are not just abstract classroom concepts — they are the hidden engines behind appreciation, depreciation, interest, and investments.

In this lesson, the focus shifts from general patterns in nature and art to patterns in finance. **Appreciation** (the increase in an asset's value over time) and **depreciation** (the decrease in value) follow predictable sequences. **Simple and compound interest** determine how savings grow or how debt accumulates. Understanding these financial sequences allows individuals to make informed decisions: Should I buy a car that loses value or a house that gains value? Is a fixed annual increase better than a percentage-based increase? How much will I really pay back on a loan?

This reading material examines how arithmetic and geometric sequences apply to real-life financial problems, including interest on loans and investments, appreciation of property, depreciation of assets, and periodic savings. It also explores how series (the sum of sequence terms) help calculate total payments or total accumulated value over time.

Part 1: Financial Sequences – Two Core Types

Most financial problems fall into two main categories, depending on whether money grows by a **fixed amount** each period or by a **fixed percentage** each period.

Arithmetic Sequence in Finance (Fixed Growth/Depreciation)

An arithmetic sequence adds the same number called the common difference (d) to each term to get the next term.

- **Appreciation by fixed amount:** A painting increases in value by ₱10,000 every year.
- **Depreciation by fixed amount:** A machine loses ₱5,000 of its value every year.
- **Fixed periodic savings:** You save ₱1,000 every month.

Formula for the n th term:

$$a_n = a_1 + (n-1)d$$

Where:

- a_n = value after n periods
- a_1 = initial value (or value after first period)
- d = common difference per period
- n = position of the term

Geometric Sequence in Finance (Percentage Growth/Depreciation)

A geometric sequence multiplies each term by a constant ratio (r) to get the next term.

- **Appreciation by percentage:** A house increases in value by 5% per year.
(Ratio = 1.05 because $100\% + 5\% = 105\% = 1.05$)
- **Depreciation by percentage:** A car loses 10% of its value each year.
(Ratio = 0.90 because $100\% - 10\% = 90\% = 0.90$)
- **Compound interest:** A bank account earns 3% interest compounded annually.
(Ratio = 1.03)

Formula for the n th term:

$$a_n = a_1 \cdot r^{n-1}$$

Where:

- a_n = value after n periods
- a_1 = value after the first period (principal $\times$ ratio)
- r = common ratio, i.e., $1 +$ growth rate (appreciation) or $1 -$ depreciation rate (depreciation)

Critical Insight: In financial problems, n counts **periods** (years, months, quarters, etc.). For compound interest, the period is the **conversion period** — the time between interest calculations.

Part 2: Essential Terms in Financial Mathematics

Before solving problems, it is necessary to understand the language of finance.

Term	Definition	Example
Appreciation	Increase in an asset's value over time	A lot bought for ₱500,000 appreciates 8% yearly
Depreciation	Decrease in an asset's value over time	A car bought for ₱800,000 depreciates to ₱500,000
Simple Interest	Interest computed only on the original principal	₱10,000 at 5% simple interest earns ₱500 per year
Compound Interest	Interest computed on principal AND accumulated past interest	₱10,000 at 5% compounded annually earns interest on interest
Loan	Money borrowed that must be repaid with interest	Borrowing ₱1 million from a bank

Term	Definition	Example
Mortgage	A loan used to buy real estate, with the property as collateral	A 15-year mortgage to buy a house
Accumulated Value (Future Value)	Total worth of an investment after a certain time, including growth	₱100,000 invested becomes ₱150,000 after 5 years
Conversion Period	Time between successive interest computations	Annually (1 year), quarterly (3 months), monthly (1 month)
Frequency of Conversion	Number of conversion periods in one year	Quarterly = 4 times per year
Periodic Rate	Interest rate for one conversion period	Annual rate 6% compounded monthly → periodic rate = $0.06/12 = 0.005$

Part 3: Solving Financial Problems Using Sequences

The following examples demonstrate step-by-step solutions using arithmetic and geometric sequences. Notice that **compound interest problems count conversion periods**, not years.

Example 1: Appreciation (Geometric Sequence)

Problem: A lot owner bought a piece of land for ₱500,000. The value of the land appreciates by 8% every year. What will be the value of the land after 5 years?

Step 1 – Identify the variable

Let a_n = value of the land after n years.

Step 2 – Identify what is being asked

We want the value after 5 years $\rightarrow a_5$.

Step 3 – Identify the given values

First year value: $a_1 = 500,000 \times 1.08 = 540,000$

Common ratio: $r = 1 + 0.08 = 1.08$

$n = 5$

Step 4 – Choose the correct sequence

Geometric sequence (percentage increase).

Step 5 – Apply the formula

$$a_n = a_1 \cdot r^{n-1}$$

$$\begin{aligned} a_5 &= 540,000 \times (1.08)^{5-1} = 540,000 \times (1.08)^4 \\ &= 540,000 \times 1.36048896 = 734,664.04 \end{aligned}$$

Step 6 – Final answer

The value of the land after 5 years is **₱734,664.04**.

Example 2: Depreciation (Geometric Sequence)

Problem: A cell phone was purchased for ₱32,500 in 2015. Its value depreciates annually by 16%. Find its value in 2025.

Step 1 – Identify the variable

Let a_n = value of the phone n years after 2015.

Step 2 – Identify what is being asked

2025 is 10 years after 2015 $\rightarrow a_{10}$.

Step 3 – Identify the given values

Depreciation rate = 16% $\rightarrow$ ratio $r = 1 - 0.16 = 0.84$

$$a_1 = 32,500 \times 0.84 = 27,300$$

$$n = 10$$

Step 4 – Choose the correct sequence

Geometric sequence (percentage decrease).

Step 5 – Apply the formula

$$\begin{aligned}a_{10} &= 27,300 \times (0.84)^{10-1} = 27,300 \times (0.84)^9 \\ &= 27,300 \times 0.208215 = 5,684.29\end{aligned}$$

Step 6 – Final answer

The value of the cell phone in 2025 is **₱5,684.29**.

Example 3: Compound Interest (Geometric Sequence with Conversion Periods)

Problem: ₱3,000 is placed in an account at 5% interest compounded quarterly for 5 years. How much is in the account at the end of 5 years?

Step 1 – Identify the variable

Let a_n = amount after n **conversion periods** (quarters).

Step 2 – Identify what is being asked

5 years $\times$ 4 quarters/year = 20 quarters $\rightarrow a_{20}$.

Step 3 – Identify the given values

Annual rate = 0.05 $\rightarrow$ periodic rate = $0.05 \div 4 = 0.0125$

$$r = 1 + 0.0125 = 1.0125$$

$$a_1 = 3,000 \times 1.0125 = 3,037.50$$

Step 4 – Choose the correct sequence

Geometric sequence (compound interest).

Step 5 – Apply the formula

$$\begin{aligned}a_{20} &= 3,037.50 \times (1.0125)^{20-1} = 3,037.50 \times (1.0125)^{19} \\ &= 3,037.50 \times 1.26677 = 3,846.11\end{aligned}$$

Step 6 – Final answer

The amount in the account after 5 years is **₱3,846.11**.

Example 4: Periodic Savings (Arithmetic Series)

Problem: Christina decides to save, starting with ₱100 in week 1. Each week, she increases her savings by ₱20. How much total money will Christina have saved after 52 weeks?

Step 1 – Identify the variable

Let a_n = amount saved in week n .

Step 2 – Identify what is being asked

Total savings after 52 weeks $\rightarrow a_{52}$.

Step 3 – Identify the given values

$a_1=100$, $d=20$, $n=52$

Step 4 – Find the 52nd term

$$a_{52}=a_1+(n-1)d=100+(51\times 20)=100+1,020=1,120$$

Step 6 – Final answer

Christina will have saved **₱1,120** after 52 weeks.

Example 5: Monthly Compound Interest (Geometric Sequence)

Problem: A savings account has an annual interest rate of 3% compounded monthly. Find the amount in the account 18 months after ₱5,000 is invested.

Step 1 – Identify the variable

Let a_n = amount after n months.

Step 2 – Identify what is being asked

value after 18 months $\rightarrow a_{18}$.

Step 3 – Identify the given values

$$\text{Periodic rate} = 0.03 \div 12 = 0.0025$$

$$r = 1 + 0.0025 = 1.0025$$

$$a_1 = 5,000 \times 1.0025 = 5,012.50$$

Step 4 – Choose the correct sequence

Geometric sequence.

Step 5 – Apply the formula

$$\begin{aligned} a_{18} &= 5,012.50 \times (1.0025)^{18-1} = 5,012.50 \times (1.0025)^{17} \\ &= 5,012.50 \times 1.0434 = 5,230.04 \end{aligned}$$

Step 6 – Final answer

The amount after 18 months is **₱5,230.04**.

Part 4: Series in Finance – Total Payments and Total Value

A **series** is the sum of the terms of a sequence. In financial problems, series answer questions like:

- **Total amount paid** on a loan after all installments
- **Total savings** after a period of regular deposits
- **Total accumulated value** of an investment with periodic contributions

Example: Arithmetic Series for Loan Payments

Suppose you pay ₱10,000 on a loan in the first month, and each month you pay ₱500 more than the previous month. After 12 months, the total paid is the sum of an arithmetic sequence with $a_1 = 10,000$, $d = 500$, $n = 12$.

$$a_{12} = 10,000 + (11 \times 500) = 15,500$$

$$S_n = n(a_1 + a_n) / 2$$

$$S_{12} = 12(10,000 + 15,500) / 2 = 6 \times 25,500 = 153,000$$

Total paid after 12 months = **₱153,000**.

Part 5: Distinguishing Arithmetic vs. Geometric in Finance

Feature	Arithmetic (Fixed Amount)	Geometric (Fixed Percentage)
Growth pattern	Linear	Exponential
Common difference/ratio	d (₱, units)	r (decimal factor)
Appreciation example	Value increases by ₱50,000/year	Value increases by 5%/year
Depreciation example	Value decreases by ₱10,000/year	Value decreases by 10%/year
Interest example	Simple interest (rarely uses sequences)	Compound interest
nth Term Formula	$a_n=a_1+(n-1)d$	$a_n=a_1\cdot r^{n-1}$

Key takeaway: If a financial problem says “increases by a fixed amount” or “decreases by a fixed amount,” think **arithmetic**. If it says “increases by a percent” or “decreases by a percent,” think **geometric**.

Key Terms

Term	Definition
Appreciation	Increase in the value of an asset over time
Depreciation	Decrease in the value of an asset over time
Simple Interest	Interest computed only on the original principal
Compound Interest	Interest computed on principal and accumulated past interest
Loan	Money borrowed that must be repaid with interest
Mortgage	A loan used to buy real estate, with the property as collateral
Accumulated Value	Total amount an investment is worth after a certain time
Conversion Period	Time between successive interest calculations
Frequency of Conversion	Number of conversion periods per year
Periodic Rate	Interest rate per conversion period
Arithmetic Sequence	Sequence with a constant difference between terms
Geometric Sequence	Sequence with a constant ratio between terms
Series	The sum of the terms of a sequence

Review Questions

Write your answers on a separate sheet of paper.

1. Explain the difference between appreciation and depreciation. Give one real-life example of each.
2. A car bought for ₱1,200,000 depreciates by 8% annually. What type of sequence models this situation? What is the common ratio?
3. A savings account earns 4% interest compounded annually. If ₱10,000 is deposited, what is the value of a_1 (the amount after the first year)?
4. A piece of machinery loses ₱15,000 of its value every year. If its initial value is ₱200,000, what will its value be after 6 years? (Arithmetic sequence)
5. A house appreciates by 6% each year. If it is worth ₱2,500,000 today, what will be its value after 5 years? (Geometric sequence)
6. An investor deposits ₱50,000 in an account with 5% interest compounded quarterly. How many conversion periods are there in 3 years? What is the periodic rate?
7. Maria saves ₱200 in week 1, ₱250 in week 2, ₱300 in week 3, and so on. How much does she save in week 20? What is her total savings after 20 weeks?
8. Why is a geometric sequence used for compound interest instead of an arithmetic sequence?
9. A loan of ₱500,000 is to be repaid with monthly payments that increase by a fixed amount each month. Is this an arithmetic or geometric situation? Explain.
10. A business owner expects her company's profit to grow by 10% each year. If the profit this year is ₱1,000,000, what will be the profit after 7 more years?

Answer Key

3. $a_1 = 10,000 \times (1 + 0.04) = \text{P}10,400$
4. $\text{Value} = 200,000 - (6 \times 15,000) = \text{P}110,000$
5. $a_5 = 2,500,000 \times (1.06)^5 = \text{P}3,345,563.94$
6. Conversion periods $= 3 \times 4 = 12$; Periodic rate $= 0.05 \div 4 = 0.0125$ or 1.25%
7. $a_{20} = 200 + (19 \times 50) = \text{P}1,150$; $S_{20} = 20(200 + 1,150)/2 = \text{P}13,500$
8. $a_7 = 1,000,000 \times (1.10)^7 = \text{P}1,948,717.10$